

Project 1 Live Recitation - Shared

1 Project 1 Live Recitation

The Python Way: Pythonic code, Jupyter shortcuts, Markdown cells, best practices on Jupyter

1.1 Jupyter Shortcuts

1.1.1 The Basics

- Arrow keys to jump between cells
- `Ctrl + Enter` to run cell (`Cmd` instead of `Ctrl` on Mac)
- `Shift + Enter` to run cell and advance to the next one

1.1.2 Organizing Cells

- `Enter` to enter a cell
- `Esc` to exit a cell
- `a` to insert a cell above current cell
- `b` to insert a cell below current cell
- `dd` to delete a cell (deleting a cell cannot be reverted)
- `m` to turn cell to Markdown
- `y` to turn cell to code

1.1.3 Once Inside Cell

- `Ctrl + arrows` to jump around text
- Highlight your code (`Ctrl + a`) and press `Ctrl + /` to comment out lines
- `Tab` for autocomplete
- `Shift + tab` to obtain docstring

1.2 Markdown Cells

A core feature of Jupyter, Markdown cells allow you to write documentation, analyses, and math.

1.2.1 Headings

Use multiple `#` characters to denote a heading. These will bring up different formats and numbers.

Level 4

Level 5 Level 6

1.2.2 L^AT_EX

Like with regular L^AT_EX, you can use \$ around mathematical symbols to bring them into typescript.

Symbols: α , β , σ

Fractions: $\frac{1}{2}$

Summation: $\sum_{i=1}^{10} i$

Double \$\$ uses a new line:

$$\int_1^{10} x dx$$

1.2.3 Special Characters

You can create bullets with an asterik and dash: * Top level - Next level

You can italicize words by wrapping them around one asterik: *italicize*

You can bold words by wrapping them around two asteriks: **bold**

Variables are often wrapped around two ‘ characters: **data**

To get multi-line formatting for variables, you can wrap multiple lines within three “ characters.

```
import pandas as pd
import numpy as np
```

1.3 Import Statements

Import packages tend to go at the top of a notebook for a number of reasons: * It looks much cleaner. * The user can just run all the imports once. * If a code cell has an import statement, it may take longer to run. * If an import statement is called within a function, calls to that function will take longer. * If the packages aren’t installed, the user can know from the get-go. - Using a file named **requirements.txt** is a common way for users to check which packages they need. - Users can just run **pip install -r requirements.txt** (pip is a Python library manager that comes with Python, -r means requirement)

An example of **requirements.txt** (obtained by running **pip freeze > requirements.txt**, this is a good idea if you want users to run a specific version of a package, like what we do in Codio):

```
opencv-python==4.4.0.46
scikit-image==0.18.1
```

```
[1]: import this
```

The Zen of Python, by Tim Peters

```
Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
```

Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than **right** now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!

We can also rename packages in import statements.

```
[2]: import numpy as np
```

1.4 Python

Python is a popular programming language for data science. Unlike Java, which is statically typed (checks types at compile-time), Python is dynamically typed. Moreover, there is no need for curly braces, semicolons, or parentheses. This makes Python a fantastic scripting language; in fact, you can run it on your terminal without creating any file.

1.4.1 Slicing

One of the key features in Python is slicing. In Java, doing these operations on arrays would be quite painful.

```
[3]: example_list = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h'] # notice the variable_
↳ declaration, Python uses snake_case (see PEP8 style guide)
```

```
[4]: example_list[0:3]
```

```
[4]: ['a', 'b', 'c']
```

```
[5]: example_list[:3]
```

```
[5]: ['a', 'b', 'c']
```

```
[6]: example_list[1:]
```

```
[6]: ['b', 'c', 'd', 'e', 'f', 'g', 'h']
```

```
[7]: example_list[:-1]
```

```
[7]: ['a', 'b', 'c', 'd', 'e', 'f', 'g']
```

```
[8]: example_list[::-1]
```

```
[8]: ['h', 'g', 'f', 'e', 'd', 'c', 'b', 'a']
```

1.4.2 Avoid For Loops When Possible

Vectorized code usually runs faster than for loops (including list comprehension) when dealing with a lot of data. Note that for loops may be faster sometimes, however, when there isn't much data or a specific Numpy or Pandas function takes time to initialize.

```
[9]: list_of_lists = [[10, 200, 10_000, 20, 4564651, 15461, 1561, 1561, 874, 56],  
                    [88, 500, 75_000, 100, 4894896, 48564, 156156, 156165, 4156,  
                    ↪62],  
                    [99, 64, 888, 1502, 156, 2165, 563, 188, 848, 5416]]
```

```
[10]: y = []  
      for list in list_of_lists:  
          for x in list:  
              if x > 100:  
                  y.append(x)  
y
```

```
[10]: [200,  
      10000,  
      4564651,  
      15461,  
      1561,  
      1561,  
      874,  
      500,  
      75000,  
      4894896,  
      48564,  
      156156,  
      156165,  
      4156,  
      888,  
      1502,  
      156,  
      2165,  
      563,  
      188,  
      848,  
      5416]
```

```
[11]: arr = np.array(list_of_lists).flatten() # converts list to Numpy array  
      arr[arr > 100]
```

```
[11]: array([[ 200,  10000, 4564651,  15461,   1561,   1561,   874,
           500,  75000, 4894896,  48564, 156156, 156165,  4156,
           888,   1502,    156,   2165,    563,   188,   848,
          5416])
```

1.5 Project 1 Answer Key

Some sections have been hidden to preserve the integrity of the project. The complete answer key will be included in the live recitation session.

```
[12]: #Data Wrangling
import numpy as np
import pandas as pd

#Plotting
import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

1.5.1 Python Help

Now that we have our imports, we want to read in the data. If you ever get stuck on what a function does, Python has a number of helpful tools.

```
[13]: # help() # interactive Python help
```

```
[14]: # help(pd.read_csv)
```

```
[15]: # pd.read_csv?
```

As you type, notice the autocomplete feature. You can also use **Shift + tab** to see the docstring signature of the function. This is really useful when you need to see which parameters are needed for a function. If you want a larger version of this, you can try **Shift + tab + tab** and keep pressing tab for larger versions.

```
[16]: college = pd.read_csv('College.csv')
```

```
[18]: # Missing code
college.head()
```

```
[18]:
```

	Private	Apps	Accept	Enroll	Top10perc	\
Names						
Abilene Christian University	Yes	1660	1232	721	23	
Adelphi University	Yes	2186	1924	512	16	
Adrian College	Yes	1428	1097	336	22	
Agnes Scott College	Yes	417	349	137	60	
Alaska Pacific University	Yes	193	146	55	16	

	Top25perc	F.Undergrad	P.Undergrad	Outstate	\
Names					
Abilene Christian University	52	2885	537	7440	
Adelphi University	29	2683	1227	12280	
Adrian College	50	1036	99	11250	
Agnes Scott College	89	510	63	12960	
Alaska Pacific University	44	249	869	7560	

	Room.Board	Books	Personal	PhD	Terminal	\
Names						
Abilene Christian University	3300	450	2200	70	78	
Adelphi University	6450	750	1500	29	30	
Adrian College	3750	400	1165	53	66	
Agnes Scott College	5450	450	875	92	97	
Alaska Pacific University	4120	800	1500	76	72	

	S.F.Ratio	perc.alumni	Expend	Grad.Rate
Names				
Abilene Christian University	18.1	12	7041	60
Adelphi University	12.2	16	10527	56
Adrian College	12.9	30	8735	54
Agnes Scott College	7.7	37	19016	59
Alaska Pacific University	11.9	2	10922	15

```
[32]: # Missing code
      many_penns
```

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
Lock Haven University of Pennsylvania	No	3570	2215	651	
Millersville University of Penn.	No	6011	3075	960	
Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
Shippensburg University of Penn.	No	5818	3281	1116	
University of Pennsylvania	Yes	12394	5232	2464	
West Chester University of Penn.	No	6502	3539	1372	
York College of Pennsylvania	Yes	2989	1855	691	

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15	55	5847	
Lock Haven University of Pennsylvania	17	41	3390	
Millersville University of Penn.	22	60	5146	
Pennsylvania State Univ. Main Campus	48	93	28938	
Shippensburg University of Penn.	14	53	5268	
University of Pennsylvania	85	100	9205	

West Chester University of Penn.	11	51	7484
York College of Pennsylvania	28	63	2988

	P.Undergrad	Outstate	Room.Board \
Names			
Bloomsburg Univ. of Pennsylvania	946	7844	2948
Lock Haven University of Pennsylvania	325	7352	3620
Millersville University of Penn.	1532	7844	3830
Pennsylvania State Univ. Main Campus	2025	10645	4060
Shippensburg University of Penn.	300	7844	3504
University of Pennsylvania	531	17020	7270
West Chester University of Penn.	1904	7844	4108
York College of Pennsylvania	1726	4990	3560

	Books	Personal	PhD	Terminal \
Names				
Bloomsburg Univ. of Pennsylvania	500	1680	66	68
Lock Haven University of Pennsylvania	225	500	47	55
Millersville University of Penn.	450	1258	72	74
Pennsylvania State Univ. Main Campus	512	2394	77	96
Shippensburg University of Penn.	450	1700	80	83
University of Pennsylvania	500	1544	95	96
West Chester University of Penn.	400	2000	76	79
York College of Pennsylvania	500	1250	75	75

	S.F.Ratio	perc.alumni	Expend \
Names			
Bloomsburg Univ. of Pennsylvania	18.0	19	7041
Lock Haven University of Pennsylvania	16.1	14	6374
Millersville University of Penn.	16.8	20	7832
Pennsylvania State Univ. Main Campus	18.1	19	8992
Shippensburg University of Penn.	18.8	13	6719
University of Pennsylvania	6.3	38	25765
West Chester University of Penn.	15.3	16	6773
York College of Pennsylvania	18.1	28	4509

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63
Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99

1.6 Jupyter Magic

Jupyter has a number of special commands that start with the % sign. You may have seen this already with `%matplotlib inline`, which tells Python to print graphs onto the notebook (instead of as a pop-up). Some other commands are:

- `%who` - shows all the saved variables
- `%%timeit` - times your code

[33]: `%who`

```
arr      college      college_least_tuition  example_list  list
list_of_lists  many_penns      mask      mpl
np      pd      penn      phd_column      phd_column_length      plt
private_column_length  private_top10  sns
this      x      y
```

[34]: `%%timeit`
`college[college.index.str.contains("Penn")]`

481 μ s $\pm$ 4.68 μ s per loop (mean $\pm$ std. dev. of 7 runs, 1000 loops each)

[35]: `%%timeit`
`college.filter(like='Penn', axis=0)`

594 μ s $\pm$ 9.8 μ s per loop (mean $\pm$ std. dev. of 7 runs, 1000 loops each)

1.7 Pandas Data Handling

[36]: `many_penns`

[36]:

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
Lock Haven University of Pennsylvania	No	3570	2215	651	
Millersville University of Penn.	No	6011	3075	960	
Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
Shippensburg University of Penn.	No	5818	3281	1116	
University of Pennsylvania	Yes	12394	5232	2464	
West Chester University of Penn.	No	6502	3539	1372	
York College of Pennsylvania	Yes	2989	1855	691	

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15	55	5847	
Lock Haven University of Pennsylvania	17	41	3390	
Millersville University of Penn.	22	60	5146	
Pennsylvania State Univ. Main Campus	48	93	28938	
Shippensburg University of Penn.	14	53	5268	
University of Pennsylvania	85	100	9205	

West Chester University of Penn.	11	51	7484
York College of Pennsylvania	28	63	2988

	P.Undergrad	Outstate	Room.Board \
Names			
Bloomsburg Univ. of Pennsylvania	946	7844	2948
Lock Haven University of Pennsylvania	325	7352	3620
Millersville University of Penn.	1532	7844	3830
Pennsylvania State Univ. Main Campus	2025	10645	4060
Shippensburg University of Penn.	300	7844	3504
University of Pennsylvania	531	17020	7270
West Chester University of Penn.	1904	7844	4108
York College of Pennsylvania	1726	4990	3560

	Books	Personal	PhD	Terminal \
Names				
Bloomsburg Univ. of Pennsylvania	500	1680	66	68
Lock Haven University of Pennsylvania	225	500	47	55
Millersville University of Penn.	450	1258	72	74
Pennsylvania State Univ. Main Campus	512	2394	77	96
Shippensburg University of Penn.	450	1700	80	83
University of Pennsylvania	500	1544	95	96
West Chester University of Penn.	400	2000	76	79
York College of Pennsylvania	500	1250	75	75

	S.F.Ratio	perc.alumni	Expend \
Names			
Bloomsburg Univ. of Pennsylvania	18.0	19	7041
Lock Haven University of Pennsylvania	16.1	14	6374
Millersville University of Penn.	16.8	20	7832
Pennsylvania State Univ. Main Campus	18.1	19	8992
Shippensburg University of Penn.	18.8	13	6719
University of Pennsylvania	6.3	38	25765
West Chester University of Penn.	15.3	16	6773
York College of Pennsylvania	18.1	28	4509

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63
Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99

```
[37]: # Missing code
many_johns
```

```
[37]:
```

	Private	Apps	Accept	Enroll	Top10perc	Top25perc	\
Names							
John Brown University	Yes	605	405	284	24	53	
John Carroll University	Yes	2421	2109	820	27	57	
Johns Hopkins University	Yes	8474	3446	911	75	94	
Johnson State College	No	833	669	279	3	13	
Saint John's University	Yes	933	800	444	18	45	
St. John's College	Yes	323	278	122	31	51	
St. John Fisher College	Yes	1368	1064	354	19	51	

	F.Undergrad	P.Undergrad	Outstate	Room.Board	\
Names					
John Brown University	961	99	6398	3672	
John Carroll University	3168	392	11700	5550	
Johns Hopkins University	3566	1569	18800	6740	
Johnson State College	1224	345	7656	4690	
Saint John's University	1691	72	12247	4081	
St. John's College	393	4	16150	5450	
St. John Fisher College	1687	677	10570	5600	

	Books	Personal	PhD	Terminal	S.F.Ratio	\
Names						
John Brown University	400	1350	68	68	13.3	
John Carroll University	600	450	89	90	14.5	
Johns Hopkins University	500	1040	96	97	3.3	
Johnson State College	500	624	80	91	14.4	
Saint John's University	500	600	76	85	12.0	
St. John's College	275	800	63	72	7.2	
St. John Fisher College	400	800	86	81	14.5	

	perc.alumni	Expend	Grad.Rate
Names			
John Brown University	19	8118	75
John Carroll University	28	7738	89
Johns Hopkins University	38	56233	90
Johnson State College	15	6564	36
Saint John's University	38	9853	70
St. John's College	26	15622	64
St. John Fisher College	29	7908	66

1.7.1 Merge

SQL operations work well on Pandas. The key function you will be using is `merge`.

```
[38]: many_penns.reset_index().merge(many_johns.reset_index(), how='outer',
    ↪indicator=True) # outer join gets the union of all rows
```

```
[38]:
```

	Names	Private	Apps	Accept	Enroll	\
0	Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
1	Lock Haven University of Pennsylvania	No	3570	2215	651	
2	Millersville University of Penn.	No	6011	3075	960	
3	Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
4	Shippensburg University of Penn.	No	5818	3281	1116	
5	University of Pennsylvania	Yes	12394	5232	2464	
6	West Chester University of Penn.	No	6502	3539	1372	
7	York College of Pennsylvania	Yes	2989	1855	691	
8	John Brown University	Yes	605	405	284	
9	John Carroll University	Yes	2421	2109	820	
10	Johns Hopkins University	Yes	8474	3446	911	
11	Johnson State College	No	833	669	279	
12	Saint John's University	Yes	933	800	444	
13	St. John's College	Yes	323	278	122	
14	St. John Fisher College	Yes	1368	1064	354	

	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	\
0	15	55	5847	946	7844	2948	
1	17	41	3390	325	7352	3620	
2	22	60	5146	1532	7844	3830	
3	48	93	28938	2025	10645	4060	
4	14	53	5268	300	7844	3504	
5	85	100	9205	531	17020	7270	
6	11	51	7484	1904	7844	4108	
7	28	63	2988	1726	4990	3560	
8	24	53	961	99	6398	3672	
9	27	57	3168	392	11700	5550	
10	75	94	3566	1569	18800	6740	
11	3	13	1224	345	7656	4690	
12	18	45	1691	72	12247	4081	
13	31	51	393	4	16150	5450	
14	19	51	1687	677	10570	5600	

	Books	Personal	PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad.Rate	\
0	500	1680	66	68	18.0	19	7041	75	
1	225	500	47	55	16.1	14	6374	63	
2	450	1258	72	74	16.8	20	7832	71	
3	512	2394	77	96	18.1	19	8992	63	
4	450	1700	80	83	18.8	13	6719	72	
5	500	1544	95	96	6.3	38	25765	93	
6	400	2000	76	79	15.3	16	6773	52	
7	500	1250	75	75	18.1	28	4509	99	
8	400	1350	68	68	13.3	19	8118	75	

9	600	450	89	90	14.5	28	7738	89
10	500	1040	96	97	3.3	38	56233	90
11	500	624	80	91	14.4	15	6564	36
12	500	600	76	85	12.0	38	9853	70
13	275	800	63	72	7.2	26	15622	64
14	400	800	86	81	14.5	29	7908	66

```

      _merge
0    left_only
1    left_only
2    left_only
3    left_only
4    left_only
5    left_only
6    left_only
7    left_only
8    right_only
9    right_only
10   right_only
11   right_only
12   right_only
13   right_only
14   right_only

```

```
[39]: many_penns.reset_index().merge(many_johns.reset_index(), how='inner',
    ↪indicator=True, on=['Books']) # on specifies the column to merge on
```

```
[39]:
```

	Names_x	Private_x	Apps_x	Accept_x	Enroll_x	\
0	Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
1	Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
2	Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
3	University of Pennsylvania	Yes	12394	5232	2464	
4	University of Pennsylvania	Yes	12394	5232	2464	
5	University of Pennsylvania	Yes	12394	5232	2464	
6	York College of Pennsylvania	Yes	2989	1855	691	
7	York College of Pennsylvania	Yes	2989	1855	691	
8	York College of Pennsylvania	Yes	2989	1855	691	
9	West Chester University of Penn.	No	6502	3539	1372	
10	West Chester University of Penn.	No	6502	3539	1372	

	Top10perc_x	Top25perc_x	F.Undergrad_x	P.Undergrad_x	Outstate_x	...	\
0	15	55	5847	946	7844	...	
1	15	55	5847	946	7844	...	
2	15	55	5847	946	7844	...	
3	85	100	9205	531	17020	...	
4	85	100	9205	531	17020	...	
5	85	100	9205	531	17020	...	

6	28	63	2988	1726	4990	...
7	28	63	2988	1726	4990	...
8	28	63	2988	1726	4990	...
9	11	51	7484	1904	7844	...
10	11	51	7484	1904	7844	...

	Outstate_y	Room.Board_y	Personal_y	PhD_y	Terminal_y	S.F.Ratio_y	\
0	18800	6740	1040	96	97	3.3	
1	7656	4690	624	80	91	14.4	
2	12247	4081	600	76	85	12.0	
3	18800	6740	1040	96	97	3.3	
4	7656	4690	624	80	91	14.4	
5	12247	4081	600	76	85	12.0	
6	18800	6740	1040	96	97	3.3	
7	7656	4690	624	80	91	14.4	
8	12247	4081	600	76	85	12.0	
9	6398	3672	1350	68	68	13.3	
10	10570	5600	800	86	81	14.5	

	perc.alumni_y	Expend_y	Grad.Rate_y	_merge
0	38	56233	90	both
1	15	6564	36	both
2	38	9853	70	both
3	38	56233	90	both
4	15	6564	36	both
5	38	9853	70	both
6	38	56233	90	both
7	15	6564	36	both
8	38	9853	70	both
9	19	8118	75	both
10	29	7908	66	both

[11 rows x 38 columns]

```
[40]: many_penns.reset_index().merge(many_johns.reset_index(), how='inner') # all of
      ↳ the rows are unique since they don't have all the same values for every cell
```

```
[40]: Empty DataFrame
Columns: [Names, Private, Apps, Accept, Enroll, Top10perc, Top25perc,
F.Undergrad, P.Undergrad, Outstate, Room.Board, Books, Personal, PhD, Terminal,
S.F.Ratio, perc.alumni, Expend, Grad.Rate]
Index: []
```

1.7.2 Concat

Concat gives the flexibility to join based on the axis (all rows or all columns).

```
[41]: pd.concat([many_penns, many_johns])
```

[41]:

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
Lock Haven University of Pennsylvania	No	3570	2215	651	
Millersville University of Penn.	No	6011	3075	960	
Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
Shippensburg University of Penn.	No	5818	3281	1116	
University of Pennsylvania	Yes	12394	5232	2464	
West Chester University of Penn.	No	6502	3539	1372	
York College of Pennsylvania	Yes	2989	1855	691	
John Brown University	Yes	605	405	284	
John Carroll University	Yes	2421	2109	820	
Johns Hopkins University	Yes	8474	3446	911	
Johnson State College	No	833	669	279	
Saint John's University	Yes	933	800	444	
St. John's College	Yes	323	278	122	
St. John Fisher College	Yes	1368	1064	354	

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15	55	5847	
Lock Haven University of Pennsylvania	17	41	3390	
Millersville University of Penn.	22	60	5146	
Pennsylvania State Univ. Main Campus	48	93	28938	
Shippensburg University of Penn.	14	53	5268	
University of Pennsylvania	85	100	9205	
West Chester University of Penn.	11	51	7484	
York College of Pennsylvania	28	63	2988	
John Brown University	24	53	961	
John Carroll University	27	57	3168	
Johns Hopkins University	75	94	3566	
Johnson State College	3	13	1224	
Saint John's University	18	45	1691	
St. John's College	31	51	393	
St. John Fisher College	19	51	1687	

	P.Undergrad	Outstate	Room.Board	\
Names				
Bloomsburg Univ. of Pennsylvania	946	7844	2948	
Lock Haven University of Pennsylvania	325	7352	3620	
Millersville University of Penn.	1532	7844	3830	
Pennsylvania State Univ. Main Campus	2025	10645	4060	
Shippensburg University of Penn.	300	7844	3504	
University of Pennsylvania	531	17020	7270	
West Chester University of Penn.	1904	7844	4108	
York College of Pennsylvania	1726	4990	3560	
John Brown University	99	6398	3672	

John Carroll University	392	11700	5550
Johns Hopkins University	1569	18800	6740
Johnson State College	345	7656	4690
Saint John's University	72	12247	4081
St. John's College	4	16150	5450
St. John Fisher College	677	10570	5600

	Books	Personal	PhD	Terminal	\
Names					
Bloomsburg Univ. of Pennsylvania	500	1680	66	68	
Lock Haven University of Pennsylvania	225	500	47	55	
Millersville University of Penn.	450	1258	72	74	
Pennsylvania State Univ. Main Campus	512	2394	77	96	
Shippensburg University of Penn.	450	1700	80	83	
University of Pennsylvania	500	1544	95	96	
West Chester University of Penn.	400	2000	76	79	
York College of Pennsylvania	500	1250	75	75	
John Brown University	400	1350	68	68	
John Carroll University	600	450	89	90	
Johns Hopkins University	500	1040	96	97	
Johnson State College	500	624	80	91	
Saint John's University	500	600	76	85	
St. John's College	275	800	63	72	
St. John Fisher College	400	800	86	81	

	S.F.Ratio	perc.alumni	Expend	\
Names				
Bloomsburg Univ. of Pennsylvania	18.0	19	7041	
Lock Haven University of Pennsylvania	16.1	14	6374	
Millersville University of Penn.	16.8	20	7832	
Pennsylvania State Univ. Main Campus	18.1	19	8992	
Shippensburg University of Penn.	18.8	13	6719	
University of Pennsylvania	6.3	38	25765	
West Chester University of Penn.	15.3	16	6773	
York College of Pennsylvania	18.1	28	4509	
John Brown University	13.3	19	8118	
John Carroll University	14.5	28	7738	
Johns Hopkins University	3.3	38	56233	
Johnson State College	14.4	15	6564	
Saint John's University	12.0	38	9853	
St. John's College	7.2	26	15622	
St. John Fisher College	14.5	29	7908	

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63

Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99
John Brown University	75
John Carroll University	89
Johns Hopkins University	90
Johnson State College	36
Saint John's University	70
St. John's College	64
St. John Fisher College	66

```
[42]: pd.concat([many_penns, many_johns], axis=1)
```

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773.0	3028.0	1025.0	
Lock Haven University of Pennsylvania	No	3570.0	2215.0	651.0	
Millersville University of Penn.	No	6011.0	3075.0	960.0	
Pennsylvania State Univ. Main Campus	No	19315.0	10344.0	3450.0	
Shippensburg University of Penn.	No	5818.0	3281.0	1116.0	
University of Pennsylvania	Yes	12394.0	5232.0	2464.0	
West Chester University of Penn.	No	6502.0	3539.0	1372.0	
York College of Pennsylvania	Yes	2989.0	1855.0	691.0	
John Brown University	NaN	NaN	NaN	NaN	
John Carroll University	NaN	NaN	NaN	NaN	
Johns Hopkins University	NaN	NaN	NaN	NaN	
Johnson State College	NaN	NaN	NaN	NaN	
Saint John's University	NaN	NaN	NaN	NaN	
St. John's College	NaN	NaN	NaN	NaN	
St. John Fisher College	NaN	NaN	NaN	NaN	

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15.0	55.0	5847.0	
Lock Haven University of Pennsylvania	17.0	41.0	3390.0	
Millersville University of Penn.	22.0	60.0	5146.0	
Pennsylvania State Univ. Main Campus	48.0	93.0	28938.0	
Shippensburg University of Penn.	14.0	53.0	5268.0	
University of Pennsylvania	85.0	100.0	9205.0	
West Chester University of Penn.	11.0	51.0	7484.0	
York College of Pennsylvania	28.0	63.0	2988.0	
John Brown University	NaN	NaN	NaN	
John Carroll University	NaN	NaN	NaN	
Johns Hopkins University	NaN	NaN	NaN	

Johnson State College	NaN	NaN	NaN
Saint John's University	NaN	NaN	NaN
St. John's College	NaN	NaN	NaN
St. John Fisher College	NaN	NaN	NaN

	P.Undergrad	Outstate	Room.Board	...	\
Names				...	
Bloomsburg Univ. of Pennsylvania	946.0	7844.0	2948.0	...	
Lock Haven University of Pennsylvania	325.0	7352.0	3620.0	...	
Millersville University of Penn.	1532.0	7844.0	3830.0	...	
Pennsylvania State Univ. Main Campus	2025.0	10645.0	4060.0	...	
Shippensburg University of Penn.	300.0	7844.0	3504.0	...	
University of Pennsylvania	531.0	17020.0	7270.0	...	
West Chester University of Penn.	1904.0	7844.0	4108.0	...	
York College of Pennsylvania	1726.0	4990.0	3560.0	...	
John Brown University	NaN	NaN	NaN	...	
John Carroll University	NaN	NaN	NaN	...	
Johns Hopkins University	NaN	NaN	NaN	...	
Johnson State College	NaN	NaN	NaN	...	
Saint John's University	NaN	NaN	NaN	...	
St. John's College	NaN	NaN	NaN	...	
St. John Fisher College	NaN	NaN	NaN	...	

	Outstate	Room.Board	Books	Personal	\
Names					
Bloomsburg Univ. of Pennsylvania	NaN	NaN	NaN	NaN	
Lock Haven University of Pennsylvania	NaN	NaN	NaN	NaN	
Millersville University of Penn.	NaN	NaN	NaN	NaN	
Pennsylvania State Univ. Main Campus	NaN	NaN	NaN	NaN	
Shippensburg University of Penn.	NaN	NaN	NaN	NaN	
University of Pennsylvania	NaN	NaN	NaN	NaN	
West Chester University of Penn.	NaN	NaN	NaN	NaN	
York College of Pennsylvania	NaN	NaN	NaN	NaN	
John Brown University	6398.0	3672.0	400.0	1350.0	
John Carroll University	11700.0	5550.0	600.0	450.0	
Johns Hopkins University	18800.0	6740.0	500.0	1040.0	
Johnson State College	7656.0	4690.0	500.0	624.0	
Saint John's University	12247.0	4081.0	500.0	600.0	
St. John's College	16150.0	5450.0	275.0	800.0	
St. John Fisher College	10570.0	5600.0	400.0	800.0	

	PhD	Terminal	S.F.Ratio	perc.alumni	\
Names					
Bloomsburg Univ. of Pennsylvania	NaN	NaN	NaN	NaN	
Lock Haven University of Pennsylvania	NaN	NaN	NaN	NaN	
Millersville University of Penn.	NaN	NaN	NaN	NaN	
Pennsylvania State Univ. Main Campus	NaN	NaN	NaN	NaN	

Shippensburg University of Penn.	NaN	NaN	NaN	NaN
University of Pennsylvania	NaN	NaN	NaN	NaN
West Chester University of Penn.	NaN	NaN	NaN	NaN
York College of Pennsylvania	NaN	NaN	NaN	NaN
John Brown University	68.0	68.0	13.3	19.0
John Carroll University	89.0	90.0	14.5	28.0
Johns Hopkins University	96.0	97.0	3.3	38.0
Johnson State College	80.0	91.0	14.4	15.0
Saint John's University	76.0	85.0	12.0	38.0
St. John's College	63.0	72.0	7.2	26.0
St. John Fisher College	86.0	81.0	14.5	29.0

	Expend	Grad.Rate
Names		
Bloomsburg Univ. of Pennsylvania	NaN	NaN
Lock Haven University of Pennsylvania	NaN	NaN
Millersville University of Penn.	NaN	NaN
Pennsylvania State Univ. Main Campus	NaN	NaN
Shippensburg University of Penn.	NaN	NaN
University of Pennsylvania	NaN	NaN
West Chester University of Penn.	NaN	NaN
York College of Pennsylvania	NaN	NaN
John Brown University	8118.0	75.0
John Carroll University	7738.0	89.0
Johns Hopkins University	56233.0	90.0
Johnson State College	6564.0	36.0
Saint John's University	9853.0	70.0
St. John's College	15622.0	64.0
St. John Fisher College	7908.0	66.0

[15 rows x 36 columns]

1.7.3 Append

Append is the specific case (`axis=0`, `join='outer'`) of `concat`.

```
[43]: many_penns.append(many_johns)
```

```
[43]:
```

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
Lock Haven University of Pennsylvania	No	3570	2215	651	
Millersville University of Penn.	No	6011	3075	960	
Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
Shippensburg University of Penn.	No	5818	3281	1116	
University of Pennsylvania	Yes	12394	5232	2464	
West Chester University of Penn.	No	6502	3539	1372	
York College of Pennsylvania	Yes	2989	1855	691	

John Brown University	Yes	605	405	284
John Carroll University	Yes	2421	2109	820
Johns Hopkins University	Yes	8474	3446	911
Johnson State College	No	833	669	279
Saint John's University	Yes	933	800	444
St. John's College	Yes	323	278	122
St. John Fisher College	Yes	1368	1064	354

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15	55	5847	
Lock Haven University of Pennsylvania	17	41	3390	
Millersville University of Penn.	22	60	5146	
Pennsylvania State Univ. Main Campus	48	93	28938	
Shippensburg University of Penn.	14	53	5268	
University of Pennsylvania	85	100	9205	
West Chester University of Penn.	11	51	7484	
York College of Pennsylvania	28	63	2988	
John Brown University	24	53	961	
John Carroll University	27	57	3168	
Johns Hopkins University	75	94	3566	
Johnson State College	3	13	1224	
Saint John's University	18	45	1691	
St. John's College	31	51	393	
St. John Fisher College	19	51	1687	

	P.Undergrad	Outstate	Room.Board	\
Names				
Bloomsburg Univ. of Pennsylvania	946	7844	2948	
Lock Haven University of Pennsylvania	325	7352	3620	
Millersville University of Penn.	1532	7844	3830	
Pennsylvania State Univ. Main Campus	2025	10645	4060	
Shippensburg University of Penn.	300	7844	3504	
University of Pennsylvania	531	17020	7270	
West Chester University of Penn.	1904	7844	4108	
York College of Pennsylvania	1726	4990	3560	
John Brown University	99	6398	3672	
John Carroll University	392	11700	5550	
Johns Hopkins University	1569	18800	6740	
Johnson State College	345	7656	4690	
Saint John's University	72	12247	4081	
St. John's College	4	16150	5450	
St. John Fisher College	677	10570	5600	

	Books	Personal	PhD	Terminal	\
Names					
Bloomsburg Univ. of Pennsylvania	500	1680	66	68	

Lock Haven University of Pennsylvania	225	500	47	55
Millersville University of Penn.	450	1258	72	74
Pennsylvania State Univ. Main Campus	512	2394	77	96
Shippensburg University of Penn.	450	1700	80	83
University of Pennsylvania	500	1544	95	96
West Chester University of Penn.	400	2000	76	79
York College of Pennsylvania	500	1250	75	75
John Brown University	400	1350	68	68
John Carroll University	600	450	89	90
Johns Hopkins University	500	1040	96	97
Johnson State College	500	624	80	91
Saint John's University	500	600	76	85
St. John's College	275	800	63	72
St. John Fisher College	400	800	86	81

	S.F.Ratio	perc.alumni	Expend	\
Names				
Bloomsburg Univ. of Pennsylvania	18.0	19	7041	
Lock Haven University of Pennsylvania	16.1	14	6374	
Millersville University of Penn.	16.8	20	7832	
Pennsylvania State Univ. Main Campus	18.1	19	8992	
Shippensburg University of Penn.	18.8	13	6719	
University of Pennsylvania	6.3	38	25765	
West Chester University of Penn.	15.3	16	6773	
York College of Pennsylvania	18.1	28	4509	
John Brown University	13.3	19	8118	
John Carroll University	14.5	28	7738	
Johns Hopkins University	3.3	38	56233	
Johnson State College	14.4	15	6564	
Saint John's University	12.0	38	9853	
St. John's College	7.2	26	15622	
St. John Fisher College	14.5	29	7908	

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63
Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99
John Brown University	75
John Carroll University	89
Johns Hopkins University	90
Johnson State College	36

Saint John's University	70
St. John's College	64
St. John Fisher College	66

[44]: many_penns # what happened?

[44]:

	Private	Apps	Accept	Enroll	\
Names					
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025	
Lock Haven University of Pennsylvania	No	3570	2215	651	
Millersville University of Penn.	No	6011	3075	960	
Pennsylvania State Univ. Main Campus	No	19315	10344	3450	
Shippensburg University of Penn.	No	5818	3281	1116	
University of Pennsylvania	Yes	12394	5232	2464	
West Chester University of Penn.	No	6502	3539	1372	
York College of Pennsylvania	Yes	2989	1855	691	

	Top10perc	Top25perc	F.Undergrad	\
Names				
Bloomsburg Univ. of Pennsylvania	15	55	5847	
Lock Haven University of Pennsylvania	17	41	3390	
Millersville University of Penn.	22	60	5146	
Pennsylvania State Univ. Main Campus	48	93	28938	
Shippensburg University of Penn.	14	53	5268	
University of Pennsylvania	85	100	9205	
West Chester University of Penn.	11	51	7484	
York College of Pennsylvania	28	63	2988	

	P.Undergrad	Outstate	Room.Board	\
Names				
Bloomsburg Univ. of Pennsylvania	946	7844	2948	
Lock Haven University of Pennsylvania	325	7352	3620	
Millersville University of Penn.	1532	7844	3830	
Pennsylvania State Univ. Main Campus	2025	10645	4060	
Shippensburg University of Penn.	300	7844	3504	
University of Pennsylvania	531	17020	7270	
West Chester University of Penn.	1904	7844	4108	
York College of Pennsylvania	1726	4990	3560	

	Books	Personal	PhD	Terminal	\
Names					
Bloomsburg Univ. of Pennsylvania	500	1680	66	68	
Lock Haven University of Pennsylvania	225	500	47	55	
Millersville University of Penn.	450	1258	72	74	
Pennsylvania State Univ. Main Campus	512	2394	77	96	
Shippensburg University of Penn.	450	1700	80	83	
University of Pennsylvania	500	1544	95	96	

West Chester University of Penn.	400	2000	76	79
York College of Pennsylvania	500	1250	75	75

	S.F.Ratio	perc.alumni	Expend \
Names			
Bloomsburg Univ. of Pennsylvania	18.0	19	7041
Lock Haven University of Pennsylvania	16.1	14	6374
Millersville University of Penn.	16.8	20	7832
Pennsylvania State Univ. Main Campus	18.1	19	8992
Shippensburg University of Penn.	18.8	13	6719
University of Pennsylvania	6.3	38	25765
West Chester University of Penn.	15.3	16	6773
York College of Pennsylvania	18.1	28	4509

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63
Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99

1.7.4 Groupby

[45]: many_penns

	Private	Apps	Accept	Enroll \
Names				
Bloomsburg Univ. of Pennsylvania	No	6773	3028	1025
Lock Haven University of Pennsylvania	No	3570	2215	651
Millersville University of Penn.	No	6011	3075	960
Pennsylvania State Univ. Main Campus	No	19315	10344	3450
Shippensburg University of Penn.	No	5818	3281	1116
University of Pennsylvania	Yes	12394	5232	2464
West Chester University of Penn.	No	6502	3539	1372
York College of Pennsylvania	Yes	2989	1855	691

	Top10perc	Top25perc	F.Undergrad \
Names			
Bloomsburg Univ. of Pennsylvania	15	55	5847
Lock Haven University of Pennsylvania	17	41	3390
Millersville University of Penn.	22	60	5146
Pennsylvania State Univ. Main Campus	48	93	28938
Shippensburg University of Penn.	14	53	5268

University of Pennsylvania	85	100	9205
West Chester University of Penn.	11	51	7484
York College of Pennsylvania	28	63	2988

	P.Undergrad	Outstate	Room.Board \
Names			
Bloomsburg Univ. of Pennsylvania	946	7844	2948
Lock Haven University of Pennsylvania	325	7352	3620
Millersville University of Penn.	1532	7844	3830
Pennsylvania State Univ. Main Campus	2025	10645	4060
Shippensburg University of Penn.	300	7844	3504
University of Pennsylvania	531	17020	7270
West Chester University of Penn.	1904	7844	4108
York College of Pennsylvania	1726	4990	3560

	Books	Personal	PhD	Terminal \
Names				
Bloomsburg Univ. of Pennsylvania	500	1680	66	68
Lock Haven University of Pennsylvania	225	500	47	55
Millersville University of Penn.	450	1258	72	74
Pennsylvania State Univ. Main Campus	512	2394	77	96
Shippensburg University of Penn.	450	1700	80	83
University of Pennsylvania	500	1544	95	96
West Chester University of Penn.	400	2000	76	79
York College of Pennsylvania	500	1250	75	75

	S.F.Ratio	perc.alumni	Expend \
Names			
Bloomsburg Univ. of Pennsylvania	18.0	19	7041
Lock Haven University of Pennsylvania	16.1	14	6374
Millersville University of Penn.	16.8	20	7832
Pennsylvania State Univ. Main Campus	18.1	19	8992
Shippensburg University of Penn.	18.8	13	6719
University of Pennsylvania	6.3	38	25765
West Chester University of Penn.	15.3	16	6773
York College of Pennsylvania	18.1	28	4509

	Grad.Rate
Names	
Bloomsburg Univ. of Pennsylvania	75
Lock Haven University of Pennsylvania	63
Millersville University of Penn.	71
Pennsylvania State Univ. Main Campus	63
Shippensburg University of Penn.	72
University of Pennsylvania	93
West Chester University of Penn.	52
York College of Pennsylvania	99

```
[46]: many_penns.reset_index().groupby('Private').mean() # group the dataframe by
      ↪ whether the colleges were private, then take the average of the the rows in
      ↪ each group across each column
```

```
[46]:
```

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	\
Private							
No	7998.166667	4247.0	1429.0	21.166667	58.833333	9345.5	
Yes	7691.500000	3543.5	1577.5	56.500000	81.500000	6096.5	

	P.Undergrad	Outstate	Room.Board	Books	Personal	\
Private						
No	1172.0	8228.833333	3678.333333	422.833333	1588.666667	
Yes	1128.5	11005.000000	5415.000000	500.000000	1397.000000	

	PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad.Rate
Private						
No	69.666667	75.833333	17.183333	16.833333	7288.5	66.0
Yes	85.000000	85.500000	12.200000	33.000000	15137.0	96.0

1.8 Exporting Your Notebook

At times, it may be necessary to export your notebook without giving users the ability to edit your code (e.g., to share with colleagues who do not have Python or just want a quick look at your work). Jupyter has the ability to export your notebook in many forms. From experience, these options work best: * HTML - preserves most of the Jupyter formatting, especially with Pandas dataframes, viewable from a browser * PDF - works best if you have a lot of LaTeX, but may be troublesome if your computer does not have all the LaTeX packages set up (i.e., MiKTeX)

Converting your Jupyter notebook to a .py file is also easy to do and useful for collaboration (OOP programming).